

Running the primary experiment

Project: Reproducible AI Research Study

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Reproducible environment and records

Without opening the old lesson, explain one key idea from Reproducible environment and records. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Pilot experiments

Without opening the old lesson, explain one key idea from Pilot experiments. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Research ethics and consent

Without opening the old lesson, explain one key idea from Research ethics and consent. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for Running the primary experiment. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Running the primary experiment. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply a new case independently

Twelve cases: 9 pass, 1 fail, 2 unassessed. Report same metrics.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

6. W2 Trace the worked case

Twenty eligible cases: 16 pass, 2 fail, 2 unassessed after outage. Plan reports completed rate plus coverage. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

7. W3 Audit the limit

A report omits this qualification: Following the plan and reporting deviations are required even when results contradict the hypothesis. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

8. W4 Apply a new case independently

Twelve cases:9 pass, 1 fail, 2 unassessed. Report same metrics.

My reasoning and evidence

9. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled running the primary experiment evidence sheet to Reproducible AI Research Study. Show how you can explain the mechanism and terminology of running the primary experiment. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Running the primary experiment

Run the frozen experiment. A research team freezes versions A and B, labels and scoring before a six-case primary run.

A scores [1,0,1,1,0,1]; B scores [1,1,1,0,1,1]. Both answer the same six cases. Binary entries mean 1=correct and 0=incorrect.

Paired primary case outcomes

	Case 1	Case 2	Case 3	Case 4	Case 5	Case 6
Version A	1	0	1	1	0	1
Version B	1	1	1	0	1	1

Read a cell using both its row and column labels.

Running the primary experiment

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Keep versions fixed

No case result may trigger tuning during this run.



2. Compute results

A totals 4/6; B totals 5/6.



3. Keep case-level records

Cases two and five favour B; case four favours A.

CHANGE ONE THING

Tune B after inspecting its case-four failure, then rescore the same set.

Explain the changed result using the matching lesson.

Compute A's and B's six-case counts and identify the cases that favour each. How must a B revision tuned on case four be labelled when evaluated again on the same set?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

13. Project portfolio milestone

Project: Reproducible AI Research Study

Current milestone: Prepare the demonstration. Gather reproducible instructions, evidence, access checks and remaining limitations.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Run the frozen experiment. A research team freezes versions A and B, labels and scoring before a six-case primary run.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Cases two and five favour B; case four favours A.

1. Compute results
2. Keep versions fixed
3. Keep case-level records

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Keep case-level records Cases two and five favour B; case four favours A. Running the primary experiment means following the declared protocol and retaining raw outcomes. Overall scores alone hide different failure patterns.

Practice 2: explain the deciding evidence

Changed-case practice: Tune B after inspecting its case-four failure, then rescore the same set. Which conclusion is supported by the supplied facts?

1. Label the rerun as development on exposed cases, not untouched primary evaluation.
2. This one example guarantees correct results for every future input.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Label the rerun as development on exposed cases, not untouched primary evaluation. The evaluation outcomes have now influenced the version.

Repair a missing building block

Review Reproducible environment and records (ai-7/24). Goal: Explain the mechanism and terminology of reproducible environment and records. Short practice: Trace the recorded mean for 2,4,6 and its changed result for 2,4,9. Which artefacts must accompany the output so another team can identify the exact run?. Return to this week and explain what you changed.

Review Pilot experiments (ai-7/22). Goal: Explain the mechanism and terminology of pilot experiments. Short practice: Calculate reviewer agreement on the four pilot answers. What can the disagreement improve, and why are those revised practice answers not untouched final evidence?. Return to this week and explain what you changed.

Review Research ethics and consent (ai-7/19). Goal: Explain the mechanism and terminology of research ethics and consent. Short practice: Identify which uses the drawing consent permits and excludes. Can the unrelated project reuse the drawings under this agreement, and what new authorisation would be needed?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

